

Flow Matching · arXiv 2210.02747

Stakeholder

18+3 · seat 4

Why FM for robot stacks

OT highlight · ROI via π_0 / OpenPI / BEHAVIOR

Sep 23 paper club · VLA Architectures

Spine: OT-CFM objective + path design for action statistics

Lipman et al. · Meta AI / Weizmann · pick a path, regress the VF

One-liner · the club spine



Moral for Sep 23

Next wins = path design for action statistics — not another ImageNet U-Net

Diffusion Policy showed generative chunks · π_0 put OT-CFM in the VLM · BEHAVIOR fixes Σ / N / soft-inpaint

FM = Continuous Normalizing Flow trained by Conditional Flow Matching · OT = straight displacement interpolant

Why FM for robot stacks

Stakeholder · ROI is downstream, not ImageNet tables

Scalable recipe

Regress VF on
straight OT paths
same U-Net family

Better metrics

NLL / FID / NFE
improve together
(matched arch)

Fewer steps

Straighter paths →
lower latency →
robot control Hz

Stack literacy

Required for any
flow-head VLA work
pair DP + π_0

ROI chain (what to fund / staff)

π_0 / $\pi_{0.5}$

OT-CFM action
expert on VLM



OpenPI

open stack +
flow head



BEHAVIOR

$\Sigma \cdot N=15 \cdot$
soft inpaint wins



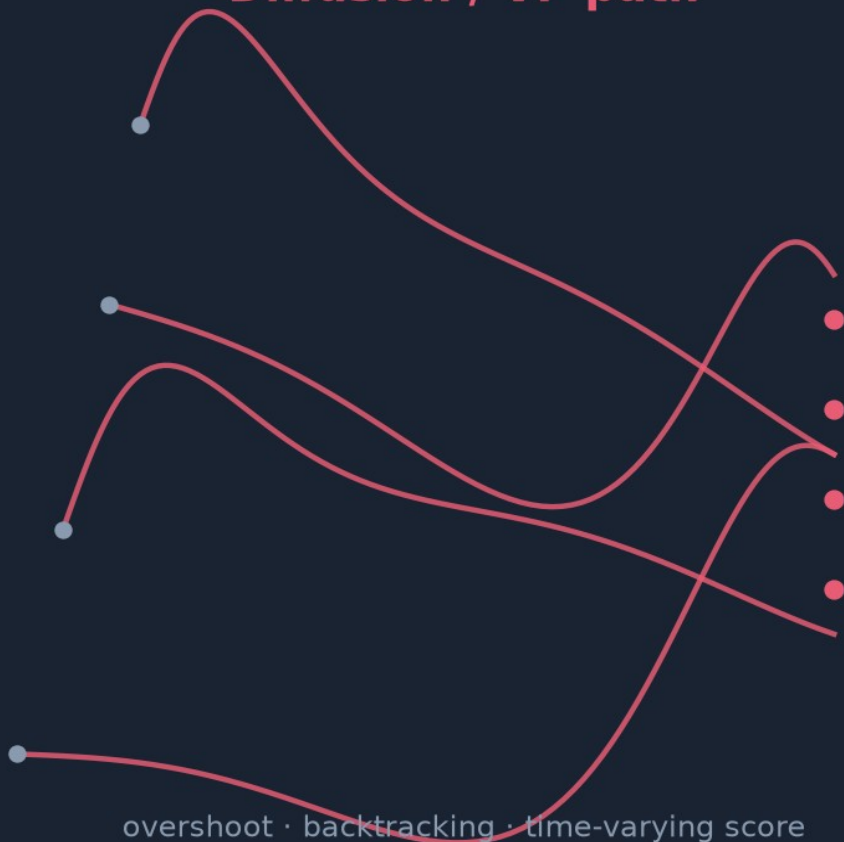
Your lab

path design +
action stats

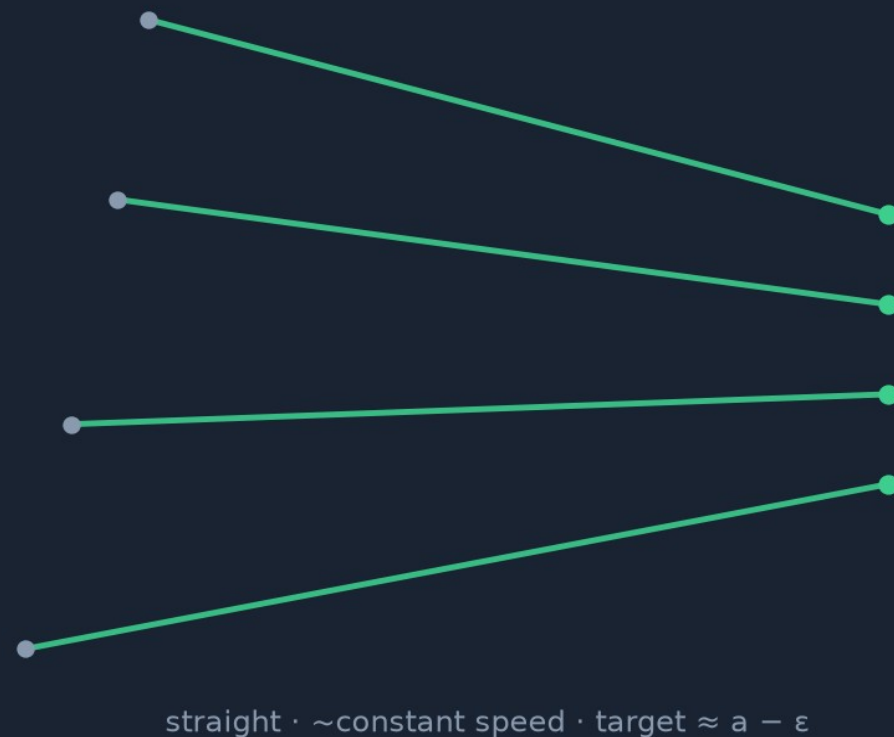
OT straight paths vs diffusion curved paths

Paper Figs 2-3 · constant-direction VF · fewer NFEs for same error

Diffusion / VP path



OT displacement path



Conditional Flow Matching · the objective card

Thm 2: $\nabla \text{CFM} = \nabla \text{FM}$ · never need the marginal VF

Intractable FM

$$L_{FM} = E_{t, p_t} \|v_t(x) - u_t(x)\|^2$$

needs marginal generating field u_t

hard — mix over all data

Tractable CFM ★

$$L_{CFM} = E_{t, q, x} \|v_t - u_t(x|x_1)\|^2$$

per-sample conditional field

same gradients (Thm 2)

OT-CFM recipe (OpenPI / π_0 action expert)

1. Sample path

$$\begin{aligned} t &\sim U(0,1) \\ \varepsilon &\sim N(0,I), \quad a \sim \text{data} \\ a_t &= (1-t)\varepsilon + t a \end{aligned}$$

2. Target VF

$$\begin{aligned} u &\approx a - \varepsilon \\ &\text{(constant direction)} \\ \sigma_{\min} &\text{softens endpoint} \end{aligned}$$

3. Regress

$$\begin{aligned} &\|v_{\theta}(a_t, t | o) - u\|^2 \\ &\text{condition on obs/VLM} \\ &\rightarrow \text{action-chunk expert} \end{aligned}$$

FLOW MATCHING FOR GENERATIVE MODELING

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ABSTRACT

We introduce a new paradigm for generative modeling built on Continuous Normalizing Flows (CNFs), allowing us to train CNFs at unprecedented scale. Specifically, we present the notion of Flow Matching (FM), a simulation-free approach for training CNFs based on regressing vector fields of fixed conditional probability paths. Flow Matching is compatible with a general family of Gaussian probability paths for transforming between noise and data samples—which subsumes existing diffusion paths as specific instances. Interestingly, we find that employing FM with diffusion paths results in a more robust and stable alternative for training diffusion models. Furthermore, Flow Matching opens the door to training CNFs with other, non-diffusion probability paths. An instance of particular interest is using Optimal Transport (OT) displacement interpolation to define the conditional probability paths. These paths are more efficient than diffusion paths, provide faster training and sampling, and result in better generalization. Training CNFs using Flow Matching on ImageNet leads to consistently better performance than alternative diffusion-based methods in terms of both likelihood and sample quality, and allows fast and reliable sample generation using off-the-shelf numerical ODE solvers.

1 INTRODUCTION

Deep generative models are a class of deep learning algorithms aimed at estimating and sampling from an unknown data distribution. The recent influx of amazing advances in generative modeling, *e.g.*, for image generation Ramesh et al. (2022); Rombach et al. (2022), is mostly facilitated by the scalable and relatively stable training of diffusion-based models Ho et al. (2020); Song et al. (2020b). However, the restriction to simple diffusion processes leads to a rather confined space of sampling probability paths, resulting in very long training times and the need to adopt specialized methods (*e.g.*, Song et al. (2020a); Zhang & Chen (2022)) for efficient sampling.

In this work we consider the general and deterministic framework of Continuous Normalizing Flows (CNFs; Chen et al. (2018)). CNFs are capable of modeling arbitrary probability path and are in particular known to encompass the probability paths modeled by diffusion processes (Song et al., 2021). However, aside from diffusion that can be trained efficiently via, *e.g.*, denoising score matching (Vincent, 2011), no scalable CNF training algorithms are known. Indeed, maximum likelihood training (*e.g.*, Grathwohl et al. (2018)) require expensive numerical ODE simulations, while existing simulation-free methods either involve intractable integrals (Rozen et al., 2021) or biased gradients (Ben-Hamu et al., 2022).

The goal of this work is to propose Flow Matching (FM), an efficient simulation-free approach to training CNF models, allowing the adoption of general probability paths to supervise CNF training. Importantly, FM breaks the barriers for scalable CNF training beyond diffusion, and sidesteps the need to reason about diffusion processes to directly work with probability paths.



Figure 1: Unconditional ImageNet-128 samples of a CNF trained using Flow Matching with Optimal Transport probability paths.

Model	CIFAR-10			ImageNet 32×32			ImageNet 64×64			Model	ImageNet 128×128	
	NLL↓	FID↓	NFE↓	NLL↓	FID↓	NFE↓	NLL↓	FID↓	NFE↓		NLL↓	FID↓
<i>Alternatives</i>												
DDPM	3.12	7.48	274	3.54	6.99	262	3.32	17.36	264	MGAN (Huang et al., 2019)	—	58.9
Score Matching	3.16	19.94	242	3.56	5.68	178	3.40	19.74	441	PacGAN2 (Song et al., 2019)	—	57.5
ScoreFlow	3.09	20.78	428	3.55	14.14	195	3.36	24.95	601	Logo-GAN-AE (Hays et al., 2019)	—	50.9
<i>Ours</i>										Self-const. GAN (Hays et al., 2019)	—	41.7
FM w/ Diffusion	3.10	8.06	183	3.54	6.37	193	3.33	16.88	187	Uncond. BigGAN (Dhariwal et al., 2019)	—	25.3
FM w/ OT	2.99	6.35	142	3.53	5.02	122	3.31	14.45	138	PGMGAN (Sachdev et al., 2020)	—	21.7
										FM w/ OT	2.90	20.9

Table 1: Likelihood (BPD), quality of generated samples (FID), and evaluation time (NFE) for the same model trained with different methods.

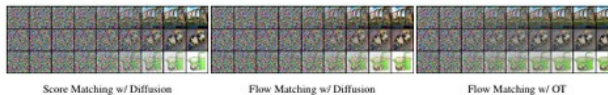


Figure 6: Sample paths from the same initial noise with models trained on ImageNet 64×64. The OT path reduces noise roughly linearly, while diffusion paths visibly remove noise only towards the end of the path. Note also the differences between the generated images.

6.1 DENSITY MODELING AND SAMPLE QUALITY ON IMAGENET

We start by comparing the same model architecture, *i.e.*, the U-Net architecture from Dhariwal & Nichol (2021) with minimal changes, trained on CIFAR-10, and ImageNet 32/64 with different popular diffusion-based losses: DDPM from (Ho et al., 2020), Score Matching (SM) (Song et al., 2020b), and Score Flow (SF) (Song et al., 2021); see Appendix E.1 for exact details. Table 1 (left) summarizes our results alongside these baselines reporting negative log-likelihood (NLL) in units of bits per dimension (BPD), sample quality as measured by the Fréchet Inception Distance (FID; Heusel et al. (2017)), and averaged number of function evaluations (NFE) required for the adaptive solver to reach its a prespecified numerical tolerance, averaged over 50k samples. All models are trained using the same architecture, hyperparameter values and number of training iterations, where baselines are allowed more iterations for better convergence. Note that these are *unconditional* models. On both CIFAR-10 and ImageNet, FM-OT consistently obtains best results across all our quantitative measures compared to competing methods. We are noticing a higher that usual FID performance in CIFAR-10 compared to previous works (Ho et al., 2020; Song et al., 2020b; 2021) that can possibly be explained by the fact that our used architecture was not optimized for CIFAR-10.

Secondly, Table 1 (right) compares a model trained using Flow Matching with the OT path on ImageNet at resolution 128×128. Our FID is state-of-the-art with the exception of IC-GAN (Casanova et al., 2021) which uses conditioning with a self-supervised ResNet50 model, and therefore is left out of this table. Figures 11, 12, 13 in the Appendix show non-curated samples from these models.

Faster training. While existing works train diffusion models with a very high number of iterations (*e.g.*, 1.3m and 10m iterations are reported by Score Flow and VDM, respectively), we find that Flow Matching generally converges much faster. Figure 5 shows FID curves during training of Flow Matching and all baselines for ImageNet 64×64; FM-OT is able to lower the FID faster and to a greater extent than the alternatives. For ImageNet-128 Dhariwal & Nichol (2021) train for 4.36m iterations with batch size 256, while FM (with 25% larger model) used 500k iterations with batch size 1.5k, *i.e.*, 33% less image throughput; see Table 3 for exact details. Furthermore, the cost of sampling from a model can drastically change during training for score matching, whereas the sampling cost stays constant when training with Flow Matching (Figure 10 in Appendix).

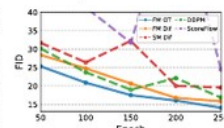


Figure 5: Image quality during training, ImageNet 64×64.

6.2 SAMPLING EFFICIENCY

For sampling, we first draw a random noise sample $x_0 \sim \mathcal{N}(0, I)$ then compute $\phi_1(x_0)$ by solving equation 1 with the trained VF, v_t , on the interval $t \in [0, 1]$ using an ODE solver. While diffusion

Downstream ROI · π_0 / OpenPI / BEHAVIOR

What Lipman gives

Simulation-free CNF training; OT paths; diffusion $\subset$ FM; ODE off-the-shelf

What DP proved

Generative continuous action chunks + receding-horizon replan work on robots

What π_0 / OpenPI do

Put OT-CFM in the VLM action expert; Lipman CLEAR cite; ~ 10 Euler steps

What BEHAVIOR adds

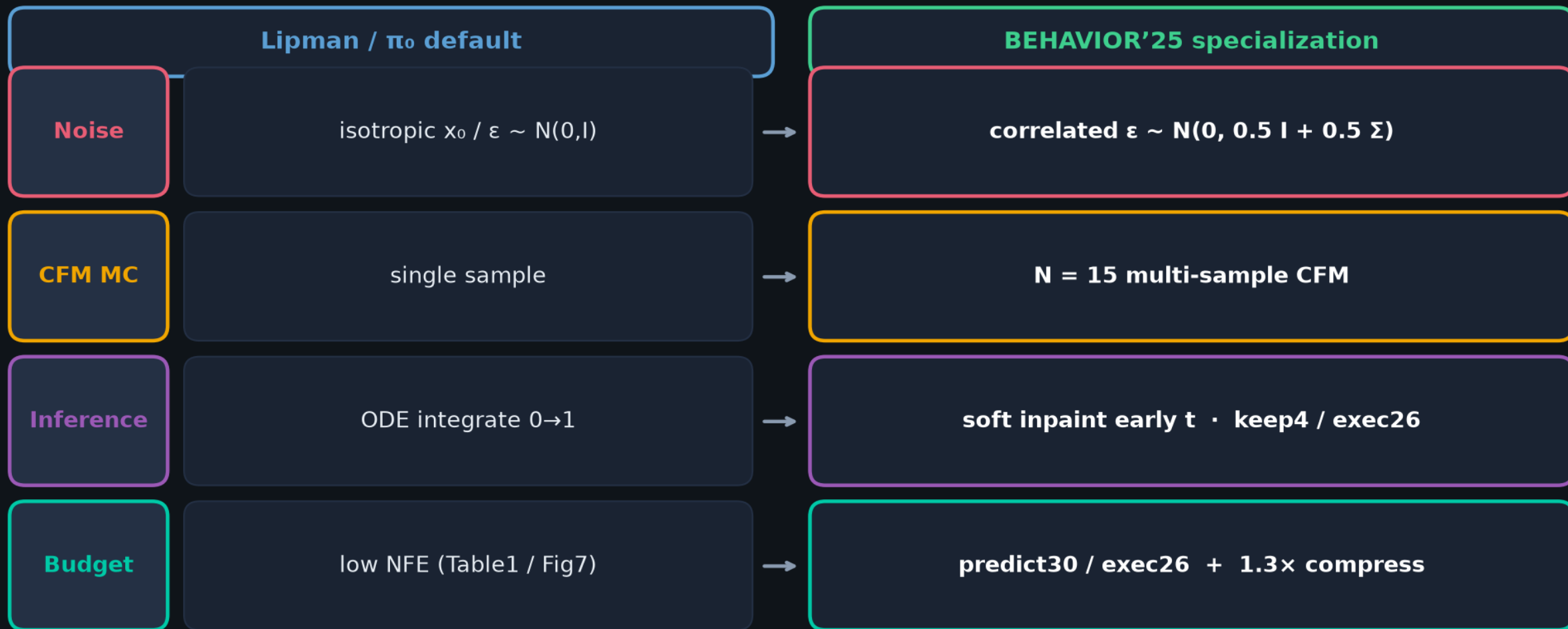
Correlated noise Σ ; $N=15$ CFM; soft inpaint keep4/exec26; $1.3\times$ compress

What you should staff

Path/ Σ design for action stats; NFE-success Pareto; not another ImageNet U-Net

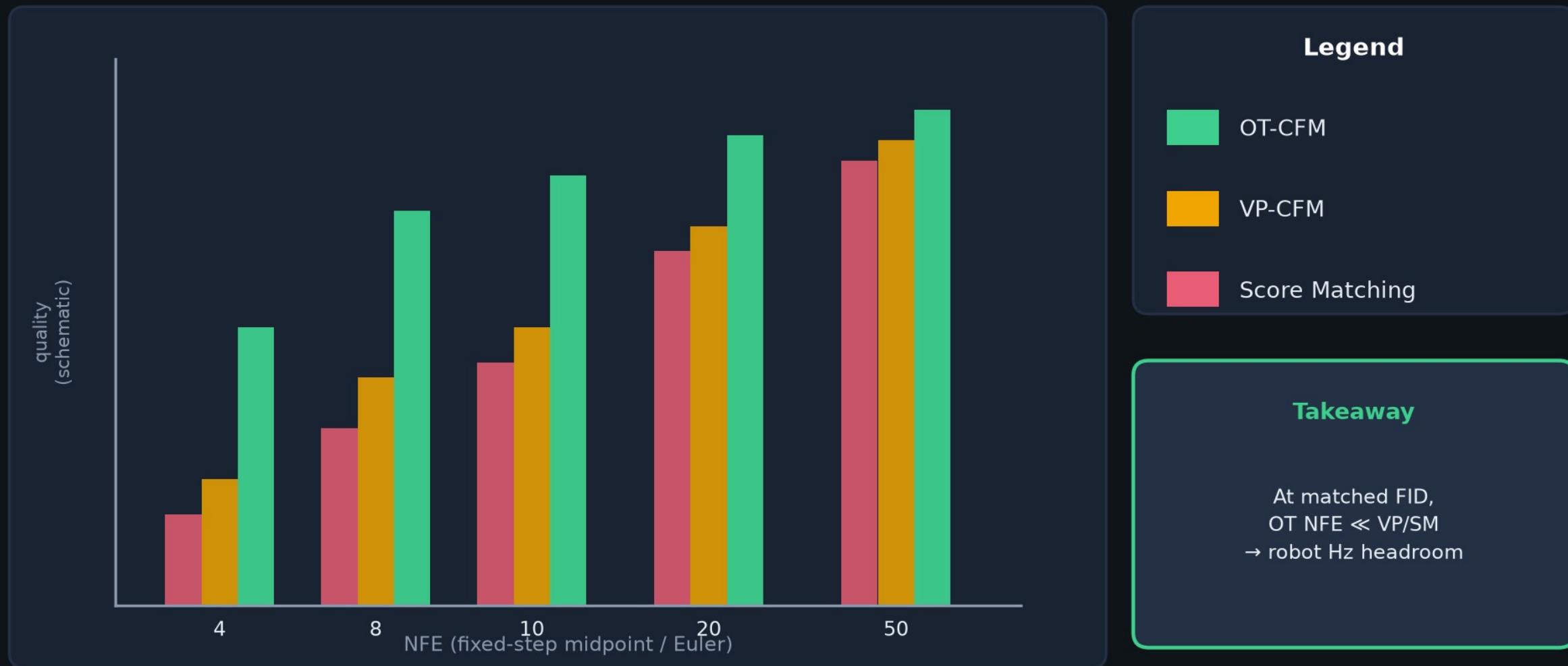
FM → BEHAVIOR map · what changed for robot stats

Lipman'22 foundation → 2025 BEHAVIOR winners on Pi0.5 / OpenPI



NFE efficiency · OT wins the low-step regime

Paper Fig 7 spirit · FM-OT ~60% NFEs of diffusion for same error · Empiricist H1



Lineage · CNF → FM → DP → π_0 → BEHAVIOR

Archaeologist spine · DP likely does NOT cite Lipman · π_0 CLEAR Lipman cite



Diffusion Policy note

Action-space DDPM chunks · conceptual bridge
likely does NOT cite Lipman · cite Chi'23 for DP

π_0 / OpenPI note

CLEAR Lipman cite · OT-style CFM action expert
 $\pi_{0.5}$ / OpenPI keep FM head · BEHAVIOR specializes

Concurrent: Rectified Flow (Liu et al.) / stochastic interpolants (Albergo & Vanden-Eijnden)
Honesty: originality is framing + scale + OT emphasis — not unique loss discovery

Don't overclaim

DP $\neq$ FM

Diffusion Policy = action-space DDPM. Conceptual bridge only. Likely no Lipman cite.

Σ / N=15 $\neq$ Lipman

Correlated noise + multi-sample CFM are BEHAVIOR specializations, not paper defaults.

IN FID $\neq$ robot KPI

ImageNet tables don't transfer to success rate / Hz / contact richness.

Conditional OT $\neq$ marginal OT

Authors admit marginal VF need not be W_2 -optimal — branding can overclaim.

Concurrent literature

Rectified Flow / interpolants overlap — cite them; claim framing+scale+OT emphasis.

8 SOCIAL RESPONSIBILITY

Along side its many positive applications, image generation can also be used for harmful proposes. Using content-controlled training sets and image validation/classification can help reduce these uses. Furthermore, the energy demand for training large deep learning models is increasing at a rapid pace (Amodei et al., 2018; Thompson et al., 2020), focusing on methods that are able to train using less gradient updates / image throughput can lead to significant time and energy savings.

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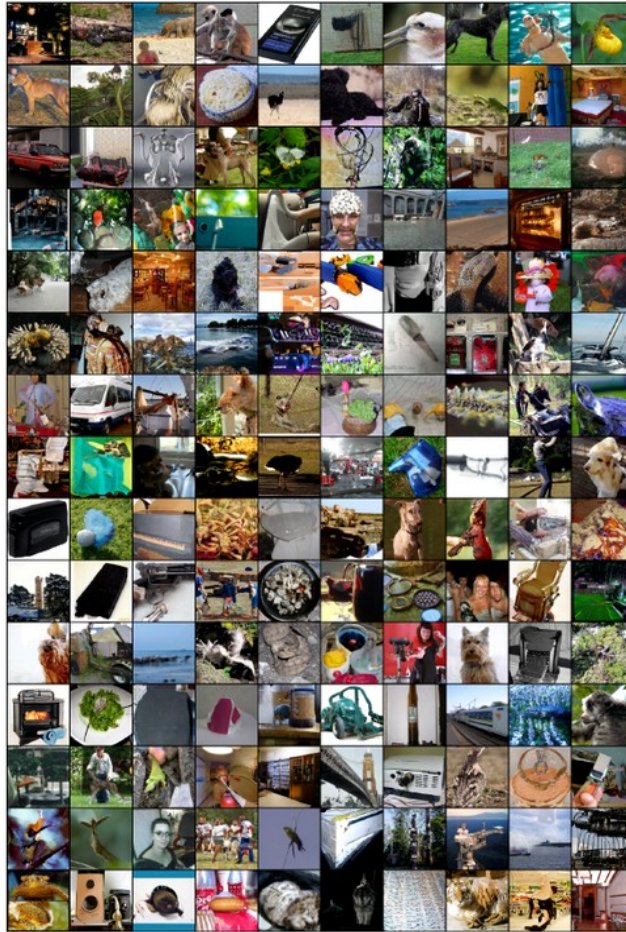


Figure 13: Non-curated unconditional ImageNet-128 generated images of a CNF trained with FM-OT.

Stakeholder · close / Q&A

Fund literacy + downstream path/ Σ work — not ImageNet bake-offs.

Spine reminder:

FM = pick a path, regress the VF; OT $\rightarrow$ straighter/faster
 $\rightarrow$ DP chunks $\rightarrow \pi_0$ OT-CFM head $\rightarrow$ BEHAVIOR fixes robot stats

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